

NITISH KUMAR SAH

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Education

National Institute of Technology, Rourkela

B.Tech in Biotechnology

Aug 2023 – Present

Rourkela, Odisha

DAV Public School, Hehal, Ranchi

CBSE, Science (PCM)

May 2022

Ranchi, Jharkhand

Relevant Coursework

- Data Structures & Algorithms
- Object Oriented Programming
- Database Management Systems
- Operating Systems
- Computer Networks
- Machine Learning

Work Experience

Saptang Labs – Computer & Network Security

Dec 2025 – April 2026

Software Engineer Intern | FastAPI, MongoDB, JWT, Redis, Celery, Docker

Remote – [Certificate](#)

- Developed **50+ secure REST API endpoints** using FastAPI with **RBAC**, **user management**, **input validation**, and **activity logging**, ensuring auditability of **100% critical user actions**.
- Designed a **multi-factor authentication system** (Email OTP + TOTP) and built an **asynchronous email pipeline** (API → Redis Queue → Worker → SMTP) with **retry handling** and **non-blocking** request flow.
- Optimized **database queries** and implemented **pagination** for high-volume APIs, reducing average response time by **30–40%** and improving large dataset handling.

Projects

NorthStar – Organizational Performance Management Platform

Ongoing

FastAPI, MongoDB, Redis, Celery, React, TypeScript, JWT

[GitHub](#) | [Live](#)

- Built a **role-based** goal management platform with **35+ FastAPI REST endpoints**, JWT authentication, RBAC, and **workflow-driven state transitions** across Employee/Manager/Admin systems.
- Engineered a **shared KPI synchronization system** for multi-user goal propagation with immutable field enforcement, approval workflows, and audit-tracked updates.
- Developed an **asynchronous** notification pipeline using **Redis + Celery**, and optimized dashboard APIs for analytics, Organisation **hierarchy traversal**, and quarterly progress computation.

CampusSentinel – AI Campus Surveillance (Runners-Up at IIT Guwahati)

Oct 2025

Django REST Framework, FastAPI, Pydantic, JWT, PostgreSQL, Redis

[GitHub](#) | [Demo](#)

- Designed a **relational PostgreSQL schema** with **12+ indexed tables** to correlate multi-source campus events.
- Built a **microservice backend** (Django REST Framework+FastAPI) with **10+ endpoints** and JWT authentication.
- Implemented **ML model integration pipelines** within FastAPI services to process and serve structured responses.

Technical Skills

- **Languages:** Go, C++, Python, SQL
- **Backend:** REST API Design, Authentication (JWT, MFA), RBAC, Input Validation
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis
- **Frameworks:** FastAPI, Django REST Framework, Pydantic, Motor, React
- **Tools:** PyCharm, WebStorm, VS Code, Docker, Celery, Git, GitHub

Achievements/Certifications

- **Competitive Programming:** CodeChef 3-Star (Rating: 1600+) [CodeChef](#)
- **Runners-Up, Ethos 2025, IIT Guwahati** [Certificate](#)
- **2nd Runners-Up, Hackfest'25, IIT-ISM Dhanbad** [Certificate](#)
- **Supervised ML: Regression and Classification** | DeepLearning.AI [Certificate](#)
- **Advanced Learning Algorithms** | DeepLearning.AI [Certificate](#)

Extracurricular Activities

- **Pantomime | Dramatics Club** | Treasurer Mar 2024 – Present
- **Webwiz | Web Development Club** | Technical Team Member Jan 2025 – Present